

Lesson 1

Troubleshooter

Lesson objectives

- View Database Usage, Services, and Pools
- View business process Queue, Usage statistics, and State information
- Resume, restart, or terminate a business process
- View system classpath and JNDI tree information
- View the cleanup process
- View environment statistics, including cache and memory used
- View Remote Perimeter Server information if present
- View Troubleshooter information about Sterling B2B Integrator cluster nodes
- Monitor node information and threads with Cluster Monitor

Troubleshooter

The Troubleshooter tool is a graphical utility for monitoring system operation and diagnosing Sterling B2B Integrator system problems.

Troubleshooter allows you to view system information on different nodes in a clustered environment, including DB usage and services, business process queue and usage information, and environment statistics

Accessing the System Troubleshooter

To access the System Troubleshooter menu:

Navigate to the Administration menu, select Operations >
System > Troubleshooter

Troubleshooting - System Status

The System Status area displays the following information:

Option	Description
Select Node	Desired cluster node to display
Start time	The date and time the system was last started or restarted
Uptime	The amount of time the system has been running since the last start or restart
Refresh Status	Refreshes active controllers
Stop the System	Stops Sterling B2B Integrator using the softstop script.
Business Process Queue Usage	Displays business process queue usage statistics
Business Process Usage	Displays business process usage statistics
Classpath Information	Displays the Sterling B2B Integrator classpath

Troubleshooting - System Status

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- **Refresh Status** - System status information is collected and reported in real time. To view the most current system status, click Refresh Status
- **Stop the System** - Stops the system using the softstop script allowing business processes to complete before stopping the system. After 5 minutes, hardstop will run. The UI will shutdown immediately.
- **Database Usage** - The top of the Database Usage page shows the Database Access Test statistics and Database Capacity statistics.

Troubleshooting - System Status

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The Database Usage page includes information about the:

- Database Access Test
 - Average Insert Time
 - Number Inserts
 - Insert Size
- Database Capacity
 - Database MB Usage
 - Database Services
 - Environmental Pool Usage

Business Process Queue Usage

The Business Process Queue Usage page enables you to diagnose problems with business process queues.

The Business Process Queue Usage page includes information about:

- Cache Usage
- Queue Statistics

Business Process Queue Usage

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The Business Process Queue Usage allows to determine:

- Number of business processes within the data size ranges that have been processed.
- Number of business processes that ran without being cached and the number that are currently in cache.
- Cache location so that you can determine the number of business processes that were found in the Soft Reference Cache, Memory Cache, and Disk Cache.

Halted or Interrupted Business Process

- A business process in a Halting, Halted, Interrupted_Man, or Interrupted_Auto state requires immediate attention because the business process has stopped processing.
- In Sterling B2B Integrator, business processes remain in a halted or interrupted state until some action is taken on the business process.
- When you notice a halted or interrupted business process, you have two options:
 - Terminate the business process.
 - Restart the business process.

Waiting Business Process

- A business process in a waiting state may not require immediate attention; the business process may be waiting on a resource or a document from another business process before completing.
- When you notice a business process in a waiting state, you have three options:
 - Allow the business process to remain in the waiting state if it is waiting on resources or a service or activity that is disabled, but will be enabled
 - Terminate the business process
 - Restart the business process

Determining the Causes

To determine the cause of an increasing number of business processes in a halting, halted, waiting, or interrupted state, review:

- The Business Process Usage report on the Operations > System > Troubleshooting page.
- Appropriate system and business process schedules to verify that they are turned on.
- The PS Report for information related to the business process execution times.

Determining the Causes

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- View the system classpath for debugging purposes and to verify whether third-party libraries are available in the classpath.
- View the system JNDI Tree for debugging purposes and verify the expected resources.

Application Status

The Application Status area of the System Troubleshooter page displays the current operational status of a Sterling B2B Integrator process environment.

Cache Usage Page

The Cache Usage page includes the following information:

Field Name	Description
Cache name	Name of the cache
Count	Number of objects in the cache
Requests	Number of times an object was requested from the cache, regardless of success
Hits	Number of times an object was requested from the cache and found

Resolutions for Inefficient Use of Cache

The following resolutions are for the inefficient use of cache:

- Improperly tuned cache performance properties
- High cache allocation for less frequently used large objects
- Low cache allocation for more frequently used small objects

Application Status - Threads

To view threads in Sterling B2B Integrator, in the Application Status area, click Threads.

Threads						
Stop	State	ID	Type	Priority	Registered	
 stop	Active	10181 [fs_3_1_whitepaper]	Harness	Queue:4 Priority:0	2/20/2006 17:06:00	
 stop	Active	10183 [DeenvelopeX12]	Harness	Queue:4 Priority:0	2/20/2006 17:06:02	
 stop	Active	10193 [DeenvelopeX12]	Harness	Queue:4 Priority:0	2/20/2006 17:06:03	
 stop	Active	10194 [DeenvelopeX12]	Harness	Queue:4 Priority:0	2/20/2006 17:06:02	
 stop	Active	10196 [DeenvelopeX12]	Harness	Queue:4 Priority:0	2/20/2006 17:06:02	
 stop	Active	10197 [DeenvelopeX12]	Harness	Queue:4 Priority:0	2/20/2006 17:06:03	
 stop	Active	10201 [fs_3_1_whitepaper]	Harness	Queue:4 Priority:0	2/20/2006 17:06:03	
 stop	Active	FileSystem_node11140473160919 [FileSystem_node1]	RMI	5	2/20/2006 17:06:00	
 stop	Active	10195 [phaethon:856d6e1:10988e89664:4300]	Business Process	Queue:4 Priority:0	2/20/2006 17:06:02	

Refreshing a Controller

User can refresh controllers running in the environment using the System Troubleshooting page.

To refresh a controller, in the Application Status area:

Click the refresh icon in the Refresh column.

Controlling Adapters

Adapters can be enabled or disabled using Troubleshooter.

Clicking the adapter name will display its service settings, characteristics, and status.

Application Status
Host: phaethon
Location: /opt/gis/node1/noapp
State: Active Memory in use: 1779709 KB Active threads: 0
[\[Cache Usage\]](#) [\[Threads\]](#)

Controllers:

Refresh	State	Name
	Active	Adapter Server
	Active	Resource Monitor Server
	Active	Scheduling Server

Adapters: 59 active / 7 stopped
21-30 of 66 Page: < 1 2 3 4 5 6 7 >

On/Off	State	Name	Type
☒	Enabled	fs_3_1_whitepaper	File System Adapter
☒	Enabled	FS_InstallValidation	File System Adapter
☒	Enabled	FS_WriteEDI	File System Adapter
☒	Enabled	FTP_Client_Adapter	FTP Client Adapter
☒	Enabled	FTP_Send_To_GIS_Support	FTP Send Adapter
☐	Stopped	FTP_Server_Adapter	FTP Server Adapter
☒	Enabled	FTPSend	FTP Send Adapter
☒	Enabled	GIS_Support_Upload_File_System_Collector	File System Adapter
☒	Enabled	HTTP_Client_Adapter	HTTP Client Adapter
☒	Enabled	HTTP_Communications_Adapter	HTTP Communications Adapter

Databases Services

In Sterling B2B Integrator, the database is installed on this server, the Database Services area displays the following information:

- Host - Name of the host on which a database resides
- Location - Location or path of the database installation
- State - State of the database, either Active or Inactive

RMIAPrime Services

The RMIAPrime Services area displays the following information:

- Host - Name of the host
- Location - Location or path to the RMIAPrime Service
- State - State of the RMIAPrime Service, either Active or Inactive

Monitoring Cluster Node Status

To view node status information navigate to:

- The Administration menu, select Operations > System > Cluster > Node Status.

Monitoring Cluster Node Status

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The following table describes the Node Status page:

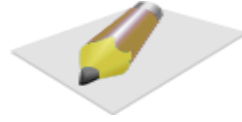
Column	Description
Name	Shows the name of the node. Click the name to view more details about the node.
URL	Shows the uniform resource locator for the node.
Troubleshooter	Provides a link to the System Troubleshooter.
Token Node	States whether this is the token node. Valid values include: True – This is the token node. False – This is not the token node.
Creation Time	Shows the date and time that the node was created.
Status	Shows the status of the node. Valid values include: Active – Node is working and available for processing Starting Ops – Node is starting up, but not available for processing Node went down – Node is not working and not available for processing

Monitoring Cluster Node Status

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Click the Name of the node to view the following detailed information:

Column	Description
Location	Shows the directory path where the node is installed.
Role	States the role of the node.
Operation Controller Host	Shows the name of the server acting as the Operation Controller Host.
Operation Controller Port	Shows the port number for the Operation Controller.
Sterling B2B Integrator version	Shows which version of Sterling B2B Integrator is used for the node.
JVM Version	Shows the version number for the java virtual machine.
JVM Vendor	Shows the vendor that provided the java virtual machine.



Exercise

1. Pool Connections

Completed lesson objectives

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Questions

